

Point-by-Point Response to Reviewers

Graph Structure Learning for Traffic Prediction

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1 General Response

Thank you for the constructive reviews of the submitted manuscript. The revision keeps the paper’s identity as a study of **graph structure learning for traffic forecasting**, with a stronger focus on **whether learned sparse graphs help because of sparsity alone or because of where edges are placed**, and on comparison with the **physical road-network adjacency**.

Two comments drove substantial experimental redesign:

- **R1-W4 (temporal interpretation of the learned DAG):** We separated contemporaneous vs. explicit multi-lag DAGMA constructions and removed the temporal-DAG claim.
- **R1-W5 (sparsity / oversmoothing / fair controls):** We added matched-edge-budget sparsity controls, structure figures, capacity checks, and a no-graph reference so that gains over the physical graph cannot be attributed only to having fewer edges.

Together these changes added substantial new experimental material, controls, and discussion (new tables, structure and control figures, sparsity sweep, temporal-resolution control, and a rewritten Limitations section). The resulting evaluation is more comprehensive and better controlled than the previous version.

Additional experiments carried out in this revision

The following experiments and evaluations are new relative to the submitted manuscript. They are described in more detail in the point-by-point responses (especially R1-W5, R1-W6, and R1-W7).

1. **Five-seed main protocol.** All primary forecasting configurations were retrained under seeds {42, 43, 44, 45, 46}; Tables 2–3 report mean RMSE with sample standard deviations, together with paired per-seed win counts (Section 4.4, “Statistical reporting”).
2. **Matched-sparsity controls.** On Los-loop PH1, RandTop30 (random directed edges), PhysTop30 (sparsified physical graph), and CorrTop30 (top training correlations) were compared with the DAGMA multi-lag graphs at the same 30-edge budget (Table 4; Figure 7).

3. **Edge-budget sweep.** The same heuristic families were evaluated for $K \in \{10, 20, 30, 50, 80\}$ directed edges (Table 5; Figure 6).
4. **Capacity control.** A graph-free T-GCN with hidden dimension 74 (parameter-matched to Mix) was trained as a single-seed check that extra gate parameters do not explain the multi-lag gain (Table 4, lower block).
5. **Temporal-resolution control.** Los-loop was resampled to 15-minute steps and the multi-lag comparison was repeated under the same five-seed protocol (Section 4.6; Table 6).
6. **Longer horizons (PH5–PH6).** The main protocol was extended to PH5 and PH6 on both native grids (Los-loop at 25–30 min; SZ-Taxi at 75–90 min), including GSL/cGSL, and is summarized in a two-panel PH1–PH6 trend figure (Appendix A.3).
7. **Limited λ_1 check.** On Los-loop PH5–PH6, contemporaneous DAGMA was refit with $\lambda_1 = 0.01$ (SZ-Taxi default) versus the Los-loop value 0.02 (Appendix A.3).
8. **Structure and qualitative figures.** Physical vs. learned multi-lag graph structure with degree histograms (Figure 5); per-seed RMSE distributions (Figure 8); predicted-vs-actual overlays on Los-loop and SZ-Taxi (Figures 9–10).
9. **Reference adjacencies.** In addition to the physical road-network graph, all main tables include an identity (no off-diagonal structure) reference so that a gain over Physical is not treated as sufficient evidence by itself.

Numbers relative to earlier drafts of this work. The evaluation pipeline and especially the DAGMA graph-learning code were substantially revised before this submission cycle. Absolute RMSE values and relative percentages therefore differ in places from any earlier draft tables. Known protocol differences that contribute to non-comparability include the five-seed reporting policy, training settings in Section 4.3 (e.g. batch size, weight decay, and the T-GCN loss with weight penalty), and the rewritten evaluation/graph-estimation code. The experimental *protocol* family (two public benchmarks, four horizons, physical baseline) is unchanged in spirit, but the revised tables are the authoritative numbers. We therefore restate the main claims from the current artifacts rather than reusing earlier percentages.

Structural changes relative to the submitted version (also recorded in the repository README):

- The submitted “Hardware specifications and hyperparameter settings” table was removed; training settings are described compactly in the Experimental Setup, and hardware/software details are documented in the repository README for reproducibility.
- Algorithm 1 (NOTEARS) and the detailed DAGMA algorithm box were removed. The revised Method cites the original papers and describes only the constructions actually used (contemporaneous and multi-lag).
- Dense multi-panel convergence grids were replaced by compact final-metric tables and a short appendix training-loss figure.

2 Reviewer 1

2.1 Strengths

Reviewer Comment

Comment 1: The core motivation — physical proximity is a poor proxy for functional traffic dependency — is well argued and nicely illustrated (Fig. 2 of the submitted version).

Response

Kept. The motivation is developed in the Introduction and Method, with the conceptual network figure (Figure 2) still showing that association need not follow geographic distance alone. We refer to **statistical associations relevant to forecasting**, not causal influence.
Location: Introduction; Section 3 (Method), motivation and Figure 2.

Reviewer Comment

Comment 2: The paragraph at the top of page 3 (motivating the shift from implicit attention weighting to explicit structure discovery) is a strong, clear articulation of the paper’s central contribution.

Response

Kept. That contrast between attention (implicit weights on a given graph) and GSL (explicit structure discovery) remains in the Introduction and is echoed in Background Section 2.4.
Location: Introduction; Section 2.4 (Graph Structure Learning).

Reviewer Comment

Comment 3: Consistent gains across two datasets, four horizons, and four metrics, with reasonable ablations and ample convergence plots.

Response

Kept and strengthened. The study still uses SZ-Taxi and Los-loop with PH1–PH4 under a common five-seed protocol. Note that absolute values differ from the previous submitted tables because the evaluation and DAGMA code were revised (see the General Response).
Location: Section 4 (Experimental Setup); Section 5 (Results), Tables 2–3.

Reviewer Comment

Comment 4: The GSL vs. cGSL asymmetry between GCN and T-GCN is a genuinely interesting empirical finding.

Response

Kept as an empirical finding; interpretation qualified. The asymmetry remains in the Results. The earlier temporal-graph reading of the contemporaneous DAG was removed. The revised Discussion offers aggregation compatibility as a possible interpretation consistent with the contrast, not as a definitive causal explanation.

Location: Section 5.2 (GSL/cGSL rows in Tables 2–3); Section 6.2 (When Learned Structure Helps).

Reviewer Comment

Comment 5: The bibliometric analysis is a thorough piece of motivating work.

Response

Kept, framed as motivation only. The bibliometric study remains as motivation (one sentence in the Introduction) with methodology and three figures in **Appendix B**.

Location: Introduction; Appendix B (Bibliometric Analysis Methodology).

2.2 Weaknesses and Questions

Reviewer Comment

Comment 6: R1-W1. The bibliometric analysis (Appendix A of the submitted version) is thorough but underused in the main text. Either lean on it harder in the Introduction to more directly motivate the GSL approach, or trim it substantially to save space for the core experimental contribution.

Response

We adopted a mixed strategy that follows both options without turning the bibliometrics into a second contribution.

1. **Lean on it in the Introduction (one load-bearing sentence).** Immediately before the paper’s central question, the Introduction now states that a bibliometric analysis of 784 traffic-forecasting articles (2000–2025) shows graph-based methods as a dominant recent focus **while many systems still construct the adjacency heuristically from road topology**, with a pointer to **Appendix B**.
2. **Keep the appendix complete but secondary.** Appendix B retains the bibliometric pipeline and **three figures**: the initial keyword network (Figure B3), word clouds (Figure B4), and the refined clustered network (Figure B5).

Location: Introduction (motivation paragraph); Appendix B.

Reviewer Comment

Comment 7: R1-W2. Sections 2.2 and 2.3 (GCN and T-GCN background) will be familiar to most readers of this venue. Suggest condensing both to a short paragraph with equation references, freeing up space for additional analysis (e.g., variance reporting, extra figures).

Response

We condensed both subsections to short, equation-referenced summaries.

1. **Removed** lengthy lists of GCN application domains and duplicated GRU/loss exposition.
2. **Kept** the standard GCN propagation equation in Section 2.2. Section 2.3 is a short prose summary of the graph-convolutional gated cell.
3. **Clarified** that the forecasting backbones used in the experiments are specified in Section 3, so Background is not read as a full architecture specification.
4. Freed space is used for **variance reporting** and **structure figures**, as recommended.

Location: Background Sections 2.2–2.3; backbones in Method Section 3.

Reviewer Comment

Comment 8: R1-W3. The notation switch in Section 3.2 ($A \rightarrow W$) is unexplained beyond a footnote. Suggest either keeping consistent notation throughout or explaining the switch in the main text rather than a footnote, right where W is introduced.

Response

We now introduce the convention **in the main text**, at the start of Section 3 (Method).

- $\mathbf{A} \in \{0, 1\}^{N \times N}$ is the **binary adjacency supplied to the forecasting backbone**.
- $W \in \mathbb{R}^{d \times d}$ is the **continuous coefficient matrix** estimated by the graph-learning procedure.
- The text states that the two symbols are not interchangeable.

W is used in the continuous-optimization subsection and in the contemporaneous and multi-lag constructions; $\mathbf{A}_{\text{GSL}}$, $\mathbf{A}_{\text{cGSL}}$, and $\mathbf{A}_k$ denote binary graphs after thresholding (and symmetrization, where relevant).

Location: Method Section 3, opening paragraphs (before Section 3.1).

Reviewer Comment

Comment 9: R1-W4. The temporal interpretation of the learned DAG in Section 5.1 is asserted rather than demonstrated. The claim that an edge $j \rightarrow i$ represents “ j at time t predicting i at time $t + 1$ ” does not obviously follow from the description in 3.2. Please clarify explicitly how the input to DAGMA is constructed, since this is load-bearing for the Section 5 discussion.

Response

We agree this was load-bearing and was not demonstrated in the previous submitted version. The revision separates two constructions and retires the claim that the contemporaneous DAG is a temporal graph.

1. **Contemporaneous GSL (Method Section 3.2).** DAGMA is fitted to simultaneous per-sensor snapshots $\mathbf{X} = \text{train_norm}[0 :: \text{PH}]$ (one row = all N sensors at the same time; **no lag blocks**). An edge is a **statistical association among present observations**; it does **not** encode “sensor j at time t predicts sensor i at time $t + 1$.”
2. **Multi-lag construction (Method Section 3.3) carries temporal organization explicitly.** Input $Z = [x(t - 3), x(t - 2), x(t - 1), x(t)]$ with $(L + 1)N$ variables; lag-specific binary blocks $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ are extracted by absolute-magnitude thresholding.
3. **Old Section 5 apparatus removed.** The submitted “spatial graph vs temporal dependency graph” paradox section is not in the revised paper. The GSL/cGSL asymmetry is discussed only as an aggregation-compatibility observation (Discussion Section 6.2).
4. **Contemporaneous vs lagged evidence (also relevant to Q4).** The two constructions are specified side by side in Sections 3.2–3.3; Results Section 5.4 compares single contemporaneous graphs with multi-lag graphs and with static-union vs lag-aligned use of the **same** multi-lag artifacts (Tables 2–3).

Location: Method Sections 3.2–3.3; Discussion Section 6.2.

Reviewer Comment

Comment 10: R1-W5. No hyperparameter optimization was performed for the baselines/learned models, and the learned graphs are likely much sparser than the physical graph. Please report the density/degree distribution of learned vs. physical graphs, and check whether some of the reported gains are attributable to reduced oversmoothing from sparsity rather than the specific learned structure. A minimal hyper-parameter sweep would help establish whether a fairer baseline narrows the gap.

Response

Together with R1-W4, this comment led to a **major redesign of the evaluation**, with the main question framed as: *does a learned sparse graph help because it is sparse, or because of how its edges are placed and used?*

In short: we report physical vs. learned graph density (Figure 5); add matched random, correlation, and sparsified-physical controls plus an edge-budget sweep on Los-loop PH1

(Tables 4–5; Figures 6–7); add a capacity check; and include a limited λ_1 check at PH5–PH6 (Appendix A.3). A full λ_1 /threshold grid is not claimed (details in (a)–(c) below).

(a) Density of physical vs learned graphs

- **Physical Los-loop:** 2626 off-diagonal edges; mean degree ≈ 12.69 .
- **Learned multi-lag union (Los-loop):** 28 edges after consumer thresholding; mean degree ≈ 0.14 .
- **Contemporaneous GSL:** 28 (Los-loop) and 8 (SZ-Taxi) directed edges at every PH.

These statistics appear in **Figure 5** (“Los-loop graph structure: physical adjacency; multi-lag union after consumer thresholding; node degrees”) and in Method Sections 3.2–3.3.

Location: Figure 5 (graph structure); Method Sections 3.2–3.3.

(b) Sparsity alone does not explain the multi-lag gain (Los-loop PH1) The paper’s primary comparison remains **learned adjacency vs. the physical road-network graph**. To separate structure quality from density and from removing the graph entirely, the evaluation adds **matched edge budgets** and a **graph-free baseline**:

1. **Graph-free baseline** (identity adjacency). On these data, dropping the physical graph often reduces error. We therefore treat a gain over Physical as necessary but not sufficient, and ask whether a learned graph also beats this control. Single contemporaneous GSL/cGSL **do not** beat the graph-free baseline on these benchmarks; multi-lag graphs with lag-aligned use on Los-loop **do**.
2. **Matched-sparsity controls at a nominal 30-edge budget (Los-loop PH1 only).** The DAGMA multi-lag construction uses 30 lag slots (lag blocks $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$); after consumer thresholding and union this yields 28 *distinct* directed edges. Heuristic controls place exactly 30 directed edges. The comparison is therefore at the same nominal budget scale, not a claim that every method has an identical final edge count after union/deduplication. On that budget scale:
 - RandTop30 (random): **worse** than the graph-free baseline (6.10 ± 0.12);
 - PhysTop30 (top physical weights, sparsified physical graph): **worse** than the graph-free baseline (5.58 ± 0.09);
 - CorrTop30 (top training correlations): **worse** than the graph-free baseline (5.39 ± 0.10);
 - DAGMA multi-lag edges with lag-aligned use (MultiGSL / Mix): **better** than the graph-free baseline (4.84 / 4.49).

Thus **sparsity alone does not explain** the multi-lag result on this cell; **edge placement and use** matter. In particular, simply keeping the strongest physical edges at the same budget does not recover the multi-lag gain. We do not claim this for all datasets or horizons.

Edge-budget sweep (new). We extended this control across $K \in \{10, 20, 30, 50, 80\}$ directed edges on Los-loop PH1 (five seeds; Table 5; Figure 6, “Edge-budget sweep on Los-loop PH1”). For every K , both RandTopK and CorrTopK remain **at or above** the graph-free baseline. At the same nominal $K = 30$ budget, the main MultiGSL (4.84) and Mix (4.49) remain well below both heuristic families and below Physical.

3. **Capacity control (single seed, labeled).** A hidden-74 graph-free model matching Mix’s parameter count stays near the hidden-64 graph-free error, so the gate’s extra parameters do not by themselves explain the Los-loop multi-lag result (Table 4, lower block).
4. **Oversmoothing language.** Dense-graph degradation is described as **consistent with** excessive neighborhood mixing or an unsuitable inductive bias; we do not claim we measured oversmoothing directly (Discussion Section 6.1).

Location:

- Section 4.5 (Matched-Sparsity and Capacity Controls).
- Section 5.3 (Is the Gain Only Sparsity?): Table 4, Table 5, Figure 6, Figure 7.
- Discussion Section 6.1; Limitations Section 6.5 (control scope).

(c) Hyperparameter optimization / fairer baseline To address the fairness concern, we added a no-graph reference and budget-matched sparse controls, including a **sparsified physical graph** (PhysTop30) and an edge-budget sweep for random/correlation graphs. PhysTop30 (5.58 ± 0.09) remains worse than the graph-free baseline, with RMSE substantially higher than MultiGSL/Mix, so a fairer sparse physical baseline does not close the multi-lag gap on Los-loop PH1. In addition, a **limited λ_1 check** was run on Los-loop PH5–PH6: refitting the contemporaneous graphs with $\lambda_1 = 0.01$ instead of 0.02 yields denser graphs but does not change the ranking relative to MultiGSL or the graph-free control (Appendix A.3). A full λ_1 /threshold grid is not claimed and remains listed as future work (Section 6.5).

Location: Section 4.5; Table 4; Appendix A.3 (λ_1 check); Section 6.5 (Limitations and Future Work).

Summary for R1-W5

Reviewer concern	How addressed
Learned graphs much sparser	Quantified (edges, mean degree) in Figure 5 and Method Sections 3.2–3.3
Gains may be oversmoothing only	No-graph reference; no credit vs Physical alone
Gains may be sparsity only	Matched nominal-30-edge Rand/Phys/Corr vs DAGMA (Los PH1; Table 4, Figure 7; sweep Table 5, Figure 6)
Fairer baseline / HPO	No-graph reference + PhysTop30 + capacity check + budget sweep; limited λ_1 check at PH5–PH6 (Appendix A.3); full grid not claimed

Reviewer Comment

Comment 11: R1-W6. No significance testing, variance, or multiple seeds are reported, despite acknowledging stochastic elements in training. Please report mean $\pm$ std over at least 3–5 seeds for the main results tables.

Response

The submitted version reported essentially single-run numbers; the revision adds repeated evaluation and a clear reporting policy.

Multiple seeds. All main forecasting configurations are trained under five seeds {42, 43, 44, 45, 46}. Each DAGMA graph is **fitted once** on the training split and then **held fixed** for those forecasting seeds, so the five runs quantify **training variability of the forecasting models**, not uncertainty from re-estimating the graph under different graph-learning seeds. We do not claim bit-level determinism of every numerical backend; the operational statement is that one stored graph artifact is reused across seeds.

Variance in the tables. The main T-GCN and GCN results tables (**Tables 2–3**) report **mean RMSE with sample standard deviations** (ddof = 1) over the five seeds. Captions note that bold marks the lowest mean, not a claim of statistical significance.

How multi-seed evidence is interpreted. Primary weight is placed on **effect size, five-seed means and spreads, and paired per-seed consistency**. For example, on Los-loop, Mix beats the graph-free baseline in 5/5 seeds at every PH1–PH4, and the graph-free baseline beats Physical in 5/5 seeds in all eight dataset×horizon cells. On SZ-Taxi, Mix and the graph-free baseline are within seed spread.

Paired *t*-tests (secondary only). For one Los-loop Mix-vs-baseline discussion we also computed paired *t*-tests on per-seed RMSE differences as a secondary diagnostic (Section 5.4; statistical-reporting paragraph of Section 4.4). With $n = 5$ these are not treated as formal proof and are not used as the primary evidence; the tables and claims rest on means, spreads, and paired win counts.

Significance language. With $n = 5$, the exact two-sided Wilcoxon signed-rank test cannot attain $p < 0.0625$. We therefore avoid definitive “statistically significant” claims and state this limitation in the statistical-reporting policy (Section 4.4) and under Limitations (Section 6.5).

A per-seed distribution figure on Los-loop (**Figure 8**, “Los-loop per-seed test RMSE for the multi-lag use ladder”) complements the tables; its means match Table 2.

Location: Section 4.4 (paragraph “Statistical reporting”); Tables 2–3; Section 5.4; Figure 8; Section 6.5.

Reviewer Comment

Comment 12: R1-W7. Results are only reported up to prediction horizon 4. Please report how performance (and the GSL/cGSL gap) evolves at longer horizons.

Response

Forecast horizons of four steps are standard in several traffic evaluations (including T-GCN and A3T-GCN), but we agree that the behavior of the methods—and of the GSL/cGSL comparison—at longer lookahead should be shown. We therefore extended the main protocol to **PH5 and PH6** and report the trend from PH1 through PH6 in the appendix.

What was run. Under the same training protocol as the main experiments (five seeds 42–46; batch 128; Adam; hidden 64; 50 epochs):

- **Los-loop** (native 5 min grid): PH5 and PH6 are 25 and 30 minutes ahead. The T-GCN family is shown for PH1–PH6 (including Mix); the GCN family is reported at PH5–PH6.
- **SZ-Taxi** (native 15 min grid): PH5 and PH6 are 75 and 90 minutes ahead. The same

variants are included for completeness, with that wall-clock caveat.

Contemporaneous GSL is re-estimated for PH5 and PH6 because its input depends on PH (26 and 30 edges on Los-loop after thresholding). Multi-lag artifacts are horizon-independent and are reused. MultiGSL-Weighted is omitted from the extension (it matches MultiGSL within 0.01 RMSE at PH1–PH4).

Where it appears. Appendix A.3 (*Trend Across Horizons, PH1–PH6*) and the two-panel trend figure (mean RMSE versus PH for Los-loop and SZ-Taxi). PH1–PH4 remain in Tables 2–3; the appendix figure is for the overall trend, not a second full results table.

What the longer horizons show.

1. **Los-loop T-GCN:** ranking is stable through PH6. T-GCN-MultiGSL-Mix has the lowest mean RMSE at every horizon; the physical graph remains the worst of the listed methods. The advantage of multi-lag use over the graph-free control **persists but shrinks**: Mix reduces RMSE by about 9–15% at PH1–PH4 and by 8.0% and 9.1% at PH5 and PH6 (ungated MultiGSL: 2.3% and 3.9% at PH5–PH6).
2. **GSL vs. cGSL (the focus of the question):** on T-GCN the two variants stay close at every horizon through PH6 (difference at most 0.06 RMSE; at PH5–PH6, $|\Delta| = 0.01$, well inside the five-seed spreads). The directed-versus-symmetrized gap does **not** widen as the horizon grows.
3. **Los-loop GCN at PH5–PH6:** the PH1–PH4 pattern continues—cGSL is the strongest learned contemporaneous variant, while MultiGSL as a static union remains worse than the physical GCN.
4. **SZ-Taxi at PH5–PH6:** learned variants stay close to the graph-free control (MultiGSL within 0.02–0.04 RMSE), consistent with the nearly empty multi-lag structure on this dataset. The Los-loop multi-lag gain does not transfer to 75–90 minutes on SZ-Taxi.

Scope. We evaluate through **PH6** on both native grids. Horizons beyond PH6 are not claimed (Section 6.5; Conclusion). The separate 15-minute Los-loop resampling control (Section 4.6; Table 6) is unchanged and remains a resolution study, not a substitute for PH5–PH6 on the native 5-minute series.

Location: Appendix A.3 (protocol, PH1–PH6 trend figure, GCN/SZ notes); Section 6.5 (horizon scope); Conclusion.

Reviewer Comment

Comment 13: R1-W8. Section 5 is somewhat repetitive and does not clearly tie back to the paper’s opening claims. Consider consolidating 5.1–5.3 into a single subsection and explicitly linking the “spatial vs. temporal graph” resolution back to the physical-vs-functional-dependency framing introduced in Section 1.

Response

The Results section has been reorganized so that each subsection answers one question and closes with a short summary that maps onto the Introduction.

Less repetition, message-oriented structure. The long, partly repetitive Results discussion of the submitted version is replaced by six subsections with the following titles:

1. Section 5.1 — *Comparison with the Physical Road-Network Graph*;
2. Section 5.2 — *Contemporaneous Learned Graphs (GSL and cGSL)*;
3. Section 5.3 — *Is the Gain Only Sparsity?*;
4. Section 5.4 — *Multi-Lag Learned Graphs*;
5. Section 5.5 — *Dataset Dependence and Temporal Resolution*;
6. Section 5.6 — *Summary of Findings*.

We reorganized Results to reduce repetition, but kept three separate subsections (5.1–5.3) rather than merging them, because each answers a distinct question (physical baseline vs. learned graphs; single learned graphs; the sparsity confound) and uses a different evidence block (main tables vs. controls tables). Merging them would either hide the sparsity control or force a long undifferentiated Results narrative. Repeating the same numeric table under three headings was avoided; 5.1 and 5.2 share Tables 2–3 without restating all cells.

Tie-back to the Introduction. The opening frames physical proximity versus data-dependent association and asks whether the adjacency should be estimated from traffic data. The primary claim is that a learned adjacency can outperform the physical road-network graph, and that on Los-loop this gain cannot be explained by sparsity alone. Section 5.1 and Section 5.4 develop that claim; Section 6.1 (“Learned Structure versus the Physical Graph”) returns to the proximity-versus-association framing explicitly. The graph-free baseline is a control used to judge the contribution of spatial structure. The old “spatial graph versus temporal dependency graph” resolution section from the submitted manuscript is not retained; that role is now played by the contemporaneous-versus-multi-lag distinction in Method and by the Discussion, without the temporal-DAG claim (see R1-W4).

Location: Sections 5.1–5.6 (Results); Section 6.1 (Discussion); Introduction (framing).

Reviewer Comment

Comment 14: R1-W9. The paper lacks an explicit limitations/weaknesses discussion. Please add a paragraph addressing scalability of DAGMA to larger road networks, sensitivity to the sparsity hyperparameter λ , and the fact that only two (relatively dated) backbones were tested.

Response

A dedicated Limitations discussion was added (Section 6.5) and covers the three issues raised, together with other scope conditions of the study.

DAGMA scalability and cost. Graph estimation with DAGMA is the heavier offline stage compared with ordinary forecasting-model training on these data. For a fixed dataset and construction, the graph is estimated **once** and reused for every forecasting seed, epoch, and test sample. Scalability to substantially larger road networks remains an open question (Section 6.5).

Meaning of λ . The reviewer’s λ refers to the DAGMA sparsity regularization coefficient λ_1 . In the revision, λ_1 is 0.02 (Los-loop) and 0.01 (SZ-Taxi) for the contemporaneous construction (Section 3.2), and 0.01 (both datasets) for the multi-lag construction (Section 3.3). Support thresholds are separate. A **limited** λ_1 check on Los-loop PH5–PH6 (0.01 versus 0.02) is reported in Appendix A.3; ranking relative to MultiGSL and the graph-free control

is unchanged. A full λ_1 /threshold grid is listed as future work (Section 6.5 and Conclusion).
Backbone coverage. The study uses GCN and T-GCN as relatively established spatial and spatiotemporal backbones, providing a controlled setting for comparing graph construction and how the graph is used. Results should not be generalized automatically to all newer attention-based, adaptive-graph, or transformer-style models; evaluating the same constructions on those architectures is a natural extension.

Other limitations (concise). Two datasets; horizons through **PH6** on the native grids (Appendix A.3; beyond PH6 not evaluated); matched-sparsity/capacity controls only on Los-loop PH1; static learned graphs (Mix changes use, not the edge set); five forecasting seeds ($n = 5$).

Location: Section 6.5 (Limitations and Future Work); Appendix A.3; Conclusion (future work: fuller λ_1 /threshold grid, horizons beyond PH6, other backbones).

Reviewer Comment

Comment 15: R1-W10. Metric definitions (Eqs. 4.1–4.4) are standard and likely unnecessary to spell out in full; consider replacing with a one-line reference and citation, using the freed space for additional qualitative results.

Response

We shortened the metrics subsection to **RMSE (primary) and MAE only** (Section 4.4), with compact formulas and a one-line definition of the evaluation set size n . Accuracy and R^2 are no longer expanded in the main text (they remain in the code base only). RMSE remains the primary metric in the main tables. There is no separate MAE appendix; additional appendix material is limited to lag ablation and training curves (Appendix A) and bibliometrics (Appendix B).

Location: Section 4.4 (Evaluation Metrics).

Reviewer Comment

Comment 16: R1-W11. Figures 5–8 (per-epoch training curves across 4 metrics $\times$ 4 horizons $\times$ 2 datasets) are dense and consume a lot of space. Consider replacing with compact bar charts of final metric values (with error bars) and moving the full curve grids to an appendix.

Response

The multi-panel per-epoch grids have been **removed from the main text**. Primary evidence is **final RMSE with sample standard deviations** (Tables 2–3; Figures 3–4 for key Los-loop PH1 configurations) plus compact structure and per-seed figures. A **log-scale training-loss figure** (two methods: T-GCN (Physical) and T-GCN-MultiGSL-Mix; Los-loop PH1 seed 42) is retained in **Appendix A.2** as an optimization diagnostic only, not as a ranking argument.

Location: Main Results tables/figures; Appendix A.2 (Training curves). Supplementary lag ablation is in Appendix A.1.

Reviewer Comment

Comment 17: R1-W12. Citation style inconsistency: several citations appear as bare “[number]” where the surrounding sentence would read more naturally as “Author et al. [number]”. Please standardize per venue style.

Response

Citations in the revised manuscript are standardized to the journal’s numbered author–date commands (`\citep` / `\citet`) throughout Background, Method, Setup, and Discussion, consistent with the `sn-mathphys-num` bibliography style used for compilation.

Location: Throughout the revised manuscript; bibliography file used at compile time.

Reviewer Comment

Comment 18: R1-Q1. Could you show a side-by-side visualization of the physical road-network graph versus the learned (GSL/cGSL) graph for one dataset segment? This would make the “physical proximity $\neq$ functional dependency” claim concrete rather than only asserted.

Response

We include a structure figure that contrasts the **physical road-network adjacency** with the **learned multi-lag union** on Los-loop, together with node-degree distributions.

Figure 5, caption “Los-loop graph structure: (a) physical adjacency (2626 off-diagonal edges; mean degree 12.69); (b) multi-lag union after consumer thresholding (28 edges; mean degree 0.14); (c) node degrees.” Captions describe these as **statistical dependency estimates**, not causal maps.

Location: Figure 5; Results Section 5.1 (Comparison with the Physical Road-Network Graph).

Reviewer Comment

Comment 19: R1-Q2. Could you add example plots of predicted vs. actual traffic speed time series for a few individual road nodes, to complement the aggregate RMSE/MAE tables with a qualitative sense of where the learned graph helps or hurts?

Response

We added qualitative figures of **predicted versus actual speeds** on Los-loop and SZ-Taxi (PH1, seed 42) for **three high-variance test nodes** over 100 consecutive test steps.

Figure 9, caption “Predicted versus actual speed on Los-loop ...”, overlays ground truth with T-GCN (Physical) and T-GCN-MultiGSL-Mix. On this dataset the separation between the two predictors is clear on the plotted nodes, and Mix tracks the true signal more closely, including sharp level changes.

Figure 10, caption “Predicted versus actual speed on SZ-Taxi ...”, uses the same layout and the same two predictors. Here as well the proposed multi-lag model is visibly closer to

the ground-truth series than the physical-graph baseline on the high-variance nodes shown. These figures are explicitly illustrative and do not replace aggregate RMSE/MAE.
Location: Figures 9–10 (Results, Multi-Lag Learned Graphs and Dataset Dependence).

Reviewer Comment

Comment 20: R1-Q3. Regarding the time-varying graph extension mentioned in the Conclusion as future work — do you have a concrete plan for how incremental graph updates would be computed (e.g., sliding-window DAGMA), and roughly what computational overhead that would add relative to the current ~ 2 -minute one-off structure learning?

Response

We thank the reviewer for raising the issue of graph-learning cost and possible incremental graph updates. In the current framework, the graph is learned once from the training data and then kept fixed during forecasting; it is not repeatedly re-estimated during inference. An incremental or sliding-window graph-learning scheme would require repeated graph estimation as new observations arrive, introducing additional computational cost. We therefore consider online graph updating an interesting direction for future work rather than part of the current forecasting pipeline.

Location: Method Sections 3.1 and 3.4; Section 6.5; Conclusion (future work).

Reviewer Comment

Comment 21: R1-Q4. On the DAG/traffic paradox discussed in Section 5: since the resolution hinges on DAGMA learning a temporal rather than spatial graph, can you provide direct evidence (e.g., the lagged data construction, or an ablation feeding genuinely simultaneous vs. lagged data into DAGMA) that supports this interpretation, rather than it being inferred post hoc from the GSL/cGSL performance split?

Response

We agree that the temporal interpretation must come from the **input construction**, not from a post-hoc reading of the GSL/cGSL split. The revision therefore separates two constructions explicitly:

1. **Contemporaneous GSL:** DAGMA on simultaneous snapshots $\mathbf{X} = \text{train_norm}[0 :: \text{PH}]$ (no lag blocks). An edge is a present-time statistical association; it is **not** interpreted as j at t predicting i at $t + 1$ (Method Section 3.2).
2. **Multi-lag:** DAGMA on $Z = [x(t - 3), \dots, x(t)]$ with lag-specific binary blocks $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ (Method Section 3.3).

Empirically, the **same multi-lag artifacts** used per timestep in T-GCN and unioned statically in GCN yield very different errors (Table 2 vs Table 3: e.g. T-GCN-MultiGSL 4.84 vs GCN-MultiGSL 9.78 on Los-loop PH1), and single contemporaneous graphs do not beat the graph-free control. This shows that **how lag-organized structure is consumed** matters; it does **not** independently validate the physical meaning of each lag edge or prove recovered

propagation delays. The old Section 5 “DAG paradox” apparatus is removed; the cGSL/GSL contrast is treated only as an aggregation-compatibility observation (see R1-W4).

Location: Method Sections 3.2–3.3; Results Section 5.4; Discussion Section 6.2; R1-W4 in this letter.

3 Reviewer 2

Reviewer Comment

Comment 22: R2-W1. Looking at the abstract, it seems like there might be a mix-up with one of the numbers. It states “up to 21.6% and 24.7% reduction in RMSE for GCN and T-GCN baselines, respectively,” but when checking against the results in Section 4.4, the 24.7% looks like it comes from the GCN average on Los-loop, not the T-GCN result. So it looks like the abstract might need to be adjusted to match what the paper actually shows.

Response

The previous submitted abstract’s “21.6% and 24.7%” figures were inconsistent with the results tables and are **removed**. The revised abstract reports a **maximum relative RMSE reduction of 43.0% versus the physical T-GCN baseline among the evaluated Los-loop horizons PH1–PH4** (i.e. the largest of the four horizon-wise reductions, not a constant 43% at every horizon), describes matched edge-budget controls that show the gain is not explained by sparsity alone, and states that a graph-free identity baseline is included as a reference control. The SZ-Taxi boundary (nearly empty multi-lag structure; much smaller improvements) is stated in the same abstract. Because the graph-estimation and training pipeline was revised, these numbers are not required to match the submitted tables. The more specific **14.5%** relative reduction versus the graph-free baseline on Los-loop PH1 (T-GCN-MultiGSL-Mix) is given in Section 5.4 and Conclusion item 3.

Location: Abstract; Results Sections 5.3–5.4; Conclusion.

Reviewer Comment

Comment 23: R2-W2. The Introduction and Conclusion talk about the method providing “explicit insights into hidden causal structure” for urban planners. That is an interesting claim, but when looking through the paper, it is hard to find support for it. There does not appear to be a visualization or heatmap of the learned matrix, and no comparison with the physical proximity graph to check if the high-weight edges match known traffic corridors.

Response

We agree that causal wording was not supported by the experiments. **All causal claims are removed** from Introduction, Method, Discussion, and Conclusion. Learned graphs are described as **thresholded, estimator-dependent statistical associations** under linear DAGMA; acyclicity is an optimization regularizer, not causal evidence (Method Section 3.2; Discussion Section 6.3).

For visualization, **Figure 5** shows the physical adjacency versus the multi-lag union on Los-

loop together with degree distributions. This is a **structural contrast** (density and degree), not a check that high-weight edges match known traffic corridors. We do **not** claim corridor recovery or causal pathways.

Location: Global claim policy; Figure 5; Method Section 3.2; Discussion Section 6.3.

Reviewer Comment

Comment 24: R2-W3. In Section 3.1, it says “the learned graph structure is static, the GRU’s temporal modeling allows the influence of each road to vary dynamically.” But then item 3 in Section 3.2’s list seems to say “Our method can adapt to changing traffic patterns over time, as the graph structure is learned from the data rather than being fixed.” These two statements do not quite line up. When looking at the code, it appears the graph is indeed computed once and stays the same.

Response

The reviewer is correct: the graph is fitted **once** and does not change over time. The contradictory “adapts to changing traffic” wording is **removed**. The revised Method states that learned adjacencies are static (Section 3.1); the Mix gate changes **how** those fixed lag graphs are used per node and timestep, not the edge set (Section 3.4).

Location: Method Sections 3.1 and 3.4; Section 6.5; Conclusion (time-varying graphs as future work).

Reviewer Comment

Comment 25: R2-W4. The symmetrisation formula for cGSL is introduced in Section 5.3, but cGSL is already being evaluated in Section 4. It might make sense to put that definition and justification earlier instead.

Response

cGSL is now defined in **Method Section 3.2** (*Contemporaneous Graph Learning (GSL and cGSL)*), immediately after GSL, as a binary symmetrization of the same stored artifact. The definition appears **before** any Results evaluation of cGSL (Section 5.2; Tables 2–3).

Location: Method Section 3.2; cGSL symmetrization equation.

Reviewer Comment

Comment 26: R2-W5. The convergence plots in Figures 5–8 are very hard to read with 16 subplots in each one. Breaking them into separate figures or maybe a summary table could help readability.

Response

The dense multi-panel per-epoch grids are **removed from the main text**. Appendix A.2 retains a log-scale training-loss figure (two methods — T-GCN (Physical) and T-GCN-MultiGSL-Mix; Los-loop PH1 seed 42) for reproducibility context only. Primary evidence is final RMSE with sample standard deviations (Tables 2–3).

Location: Appendix A.2 (Training curves); see also R1-W11 in this letter.

Reviewer Comment

Comment 27: R2-W6. On a smaller note, there is a typo in Section 4.4: “avergae”.

Response

The typo no longer appears in the rewritten Setup and Results text of the revised manuscript.

Location: Section 4.4 (Evaluation Metrics) in the revised manuscript.